Fundamental of Machine learning

MATPLOTLIB AND SKLEARN

Matplotlib

Matplotlib is an amazing visualization library

Matplotlib is a multi-platform

It allows us visual access to huge amounts of data

Matplotlib consists of several plots: **line**, **bar**, **scatter**, ..

Matplotlib

Importing **matplotlib**

- From matplotlib import pyplot as plt
- Import matplotlib.pyplot as plt

Matplotlib

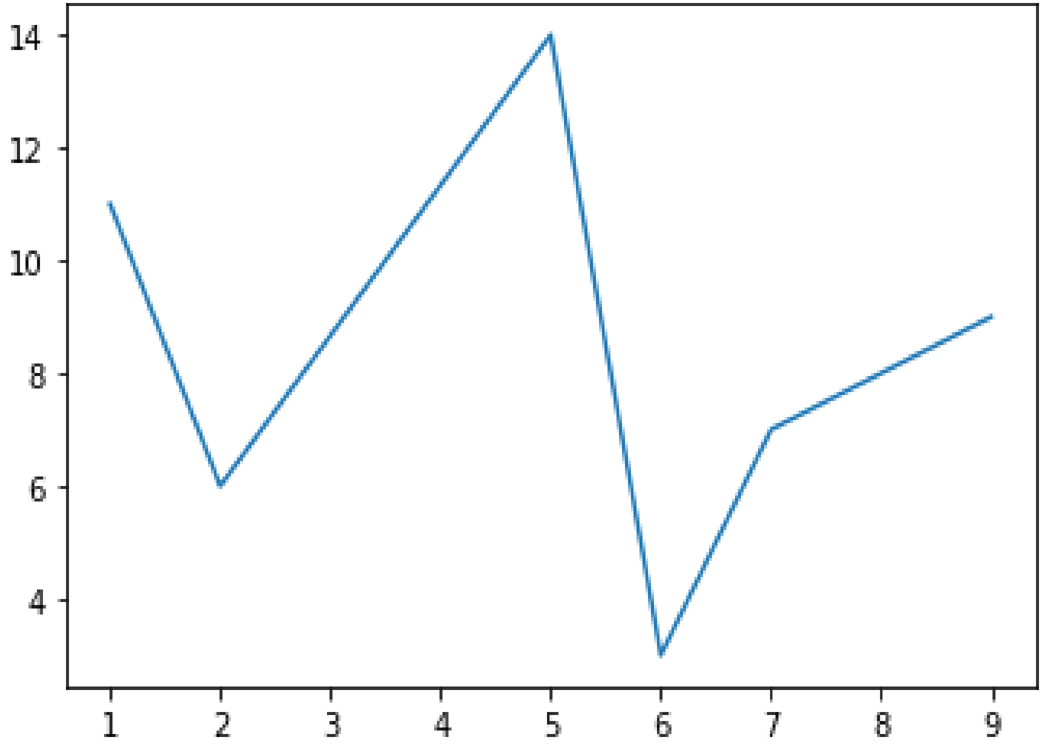
Basic plots in **Matplotlib:**

- Matplotlib comes with a wide variety of plots
- Plots helps to understand trends, patterns, and to make correlations
- They're typically instruments for reasoning about quantitative information

Matplotlib

Line plot

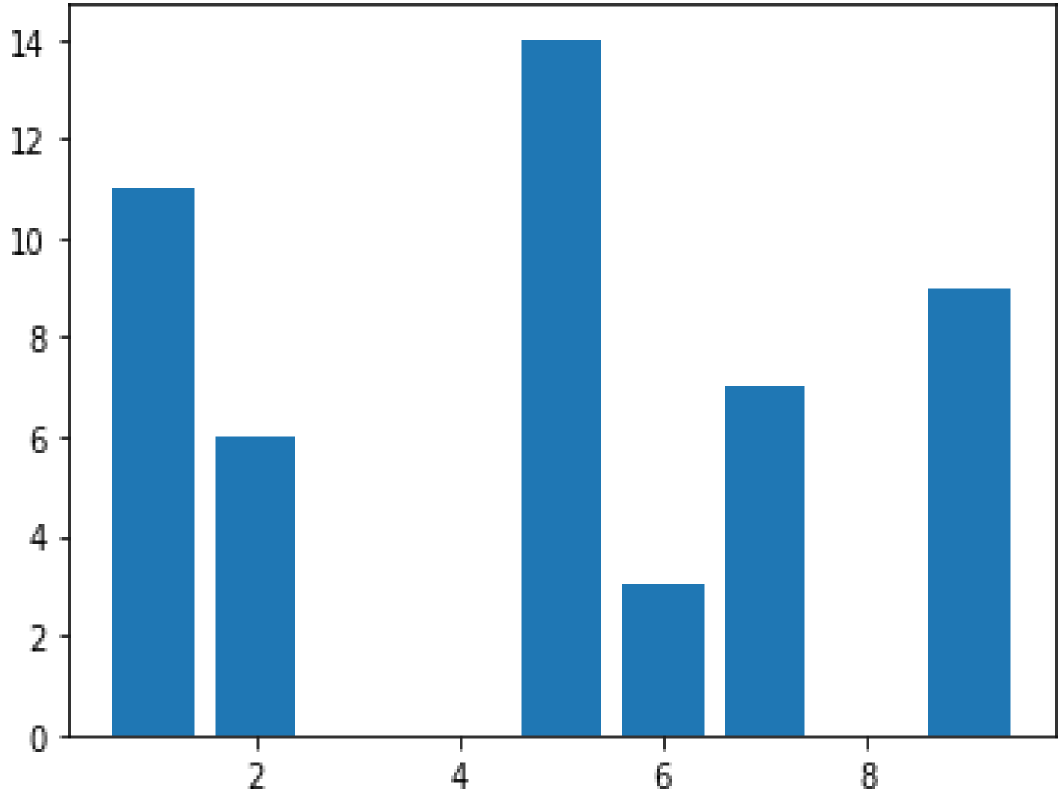
```
# Import module
from matplotlib import pyplot as plt

# x-axis values
x = [1, 2, 5, 6, 7, 9]

# y-axis values
y = [11, 6, 14, 3, 7, 9]

# Function to plot
plt.plot(x, y)
plt.show()
```



Matplotlib

Bar plot

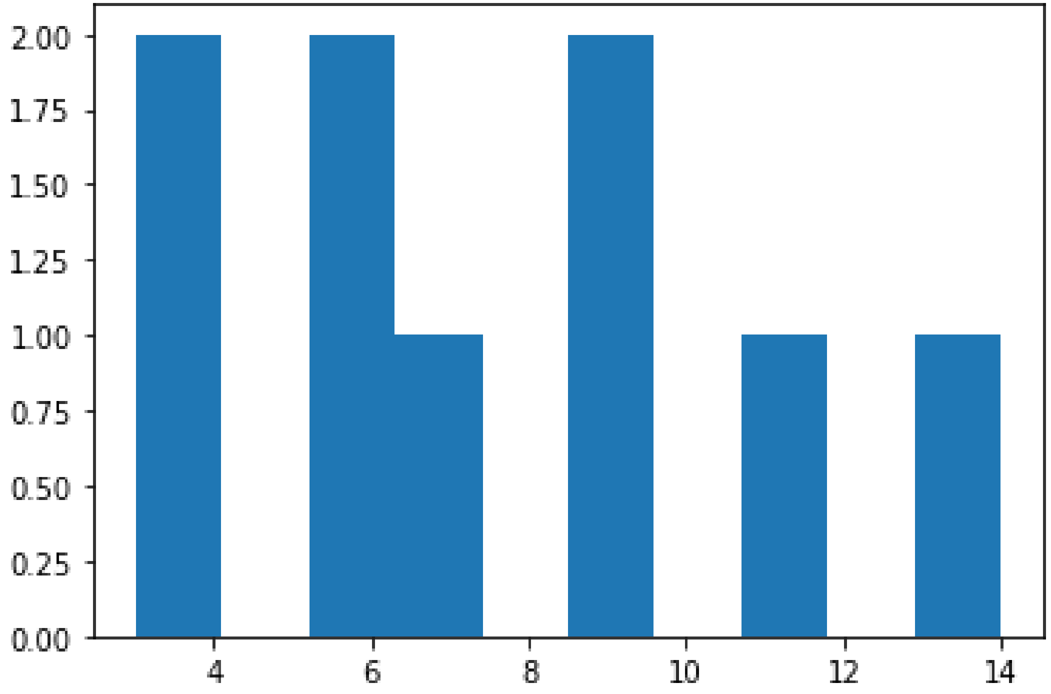
```python
# Import module
from matplotlib import pyplot as plt

# x-axis values
x = [1, 2, 5, 6, 7, 9]

# y-axis values
y = [11, 6, 14, 3, 7, 9]

# Function to plot
plt.bar(x, y)
plt.show()
```



Matplotlib

Histogram

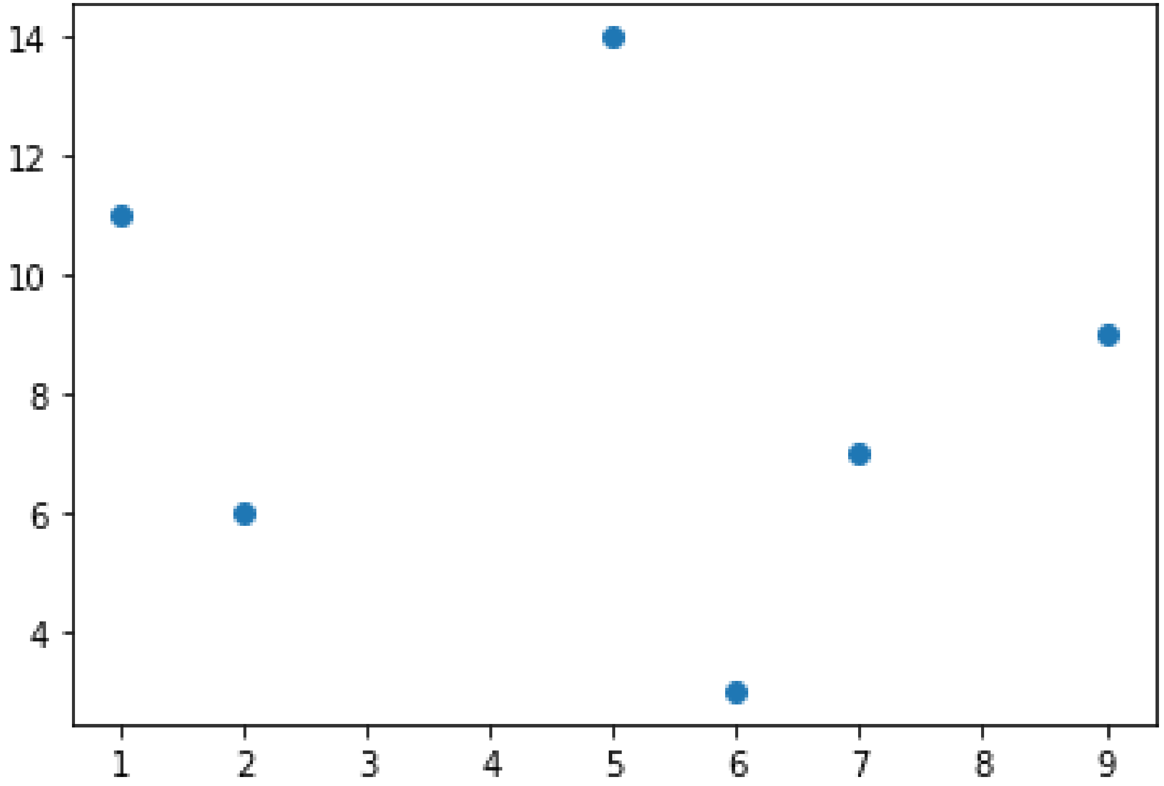
```
# Import module
from matplotlib import pyplot as plt

# y-axis values
y = [11, 6, 14, 3, 7, 9, 9, 3, 6]

# Function to plot
plt.hist(y)
plt.show()
```



Matplotlib

Scatter Plot

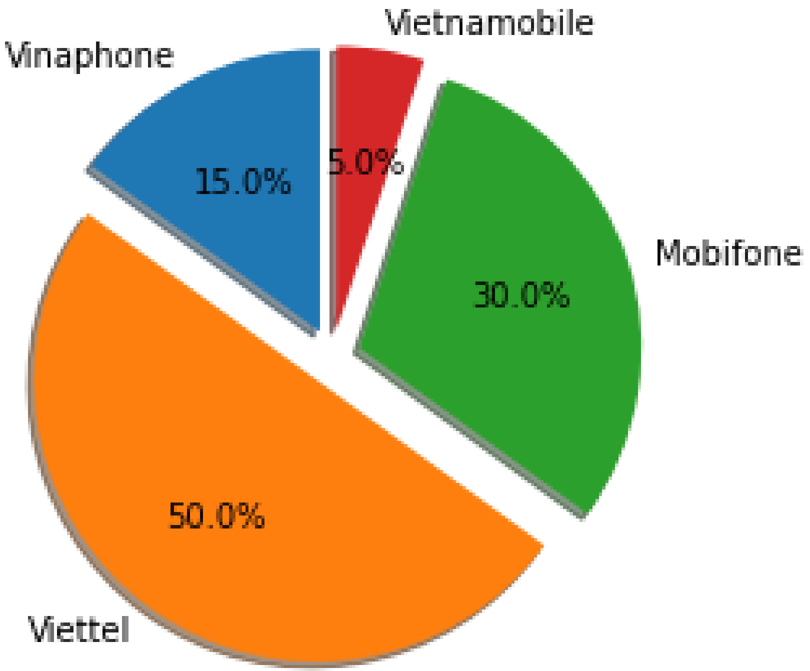
```python
# Import module
from matplotlib import pyplot as plt

# x-axis values
x = [1, 2, 5, 6, 7, 9]

# y-axis values
y = [11, 6, 14, 3, 7, 9]

# Function to plot
plt.scatter(x, y)
plt.show()
```



Matplotlib

Pie Plot

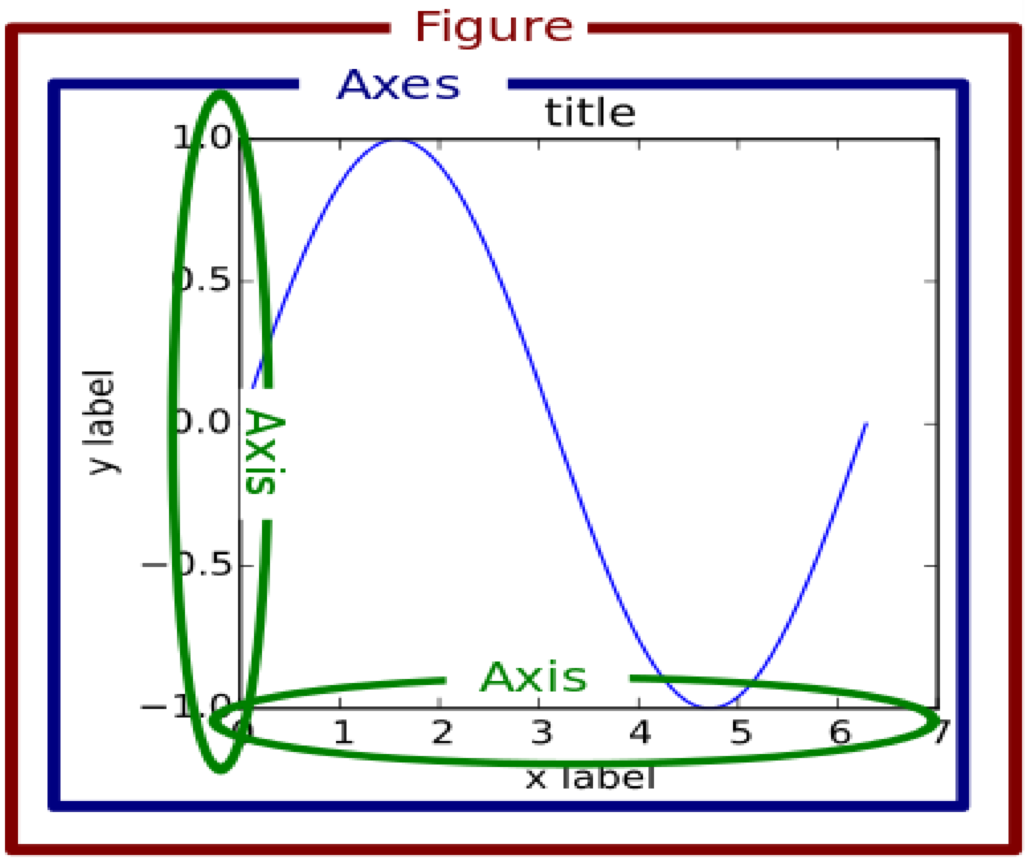
```
# Import module
from matplotlib import pyplot as plt


labels = 'Vinaphone', 'Viettel', 'Mobifone', 'V
ietnamobile'
sizes = [15, 50, 30, 5]
explode = (0.1, 0.1, 0.1, 0.1)

plt.pie(sizes, explode=explode, labels=labels,
autopct='%1.1f%%',
        shadow=True, startangle=90)

plt.show()
```



Matplotlib

The Matplotlib Object Hierarchy

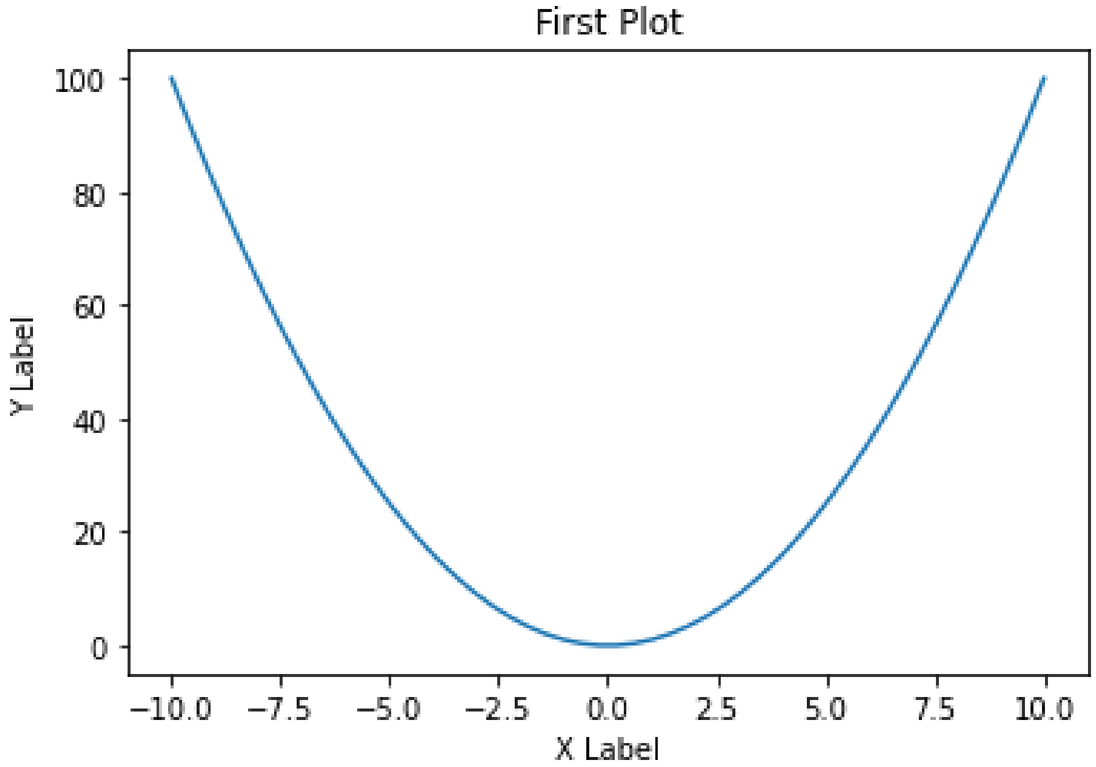
- Figure
- Axes
- Axis
- Title
- Label



Matplotlib

Example

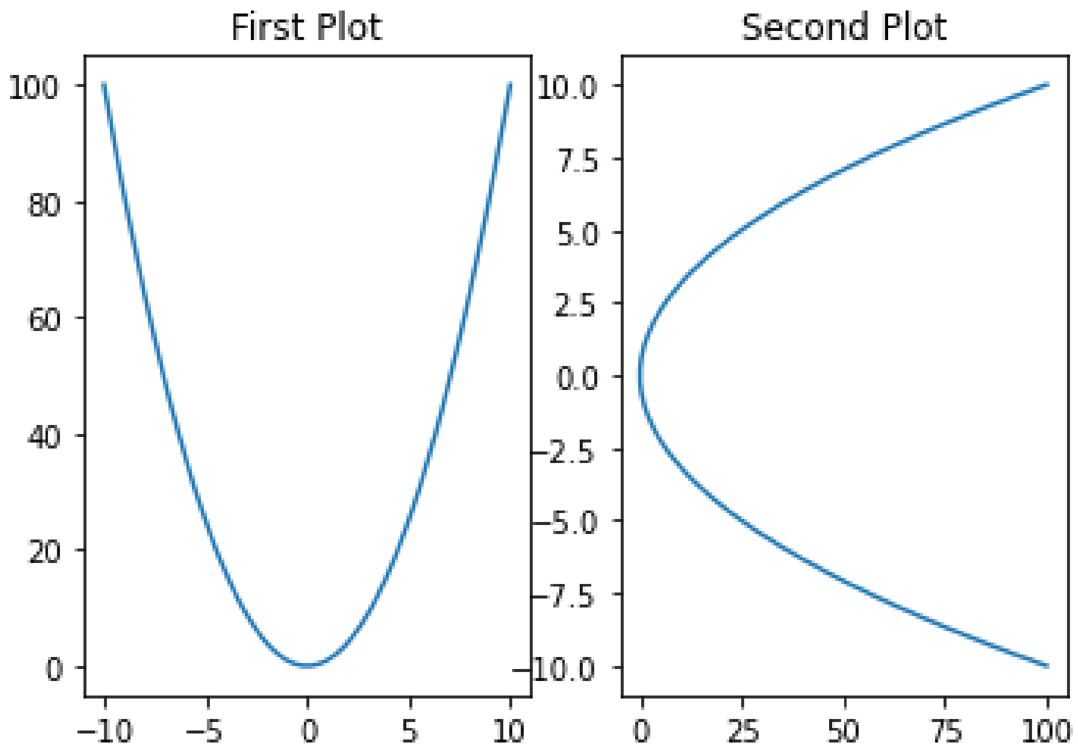
```
# Import module
from matplotlib import pyplot as plt
import numpy as np

x = np.linspace(-10, 10, 50)
y = x**2

plt.plot(x, y)
plt.title('First Plot')
plt.xlabel('X Label')
plt.ylabel('Y Label')
plt.show()
```



Matplotlib

Multi-plots:

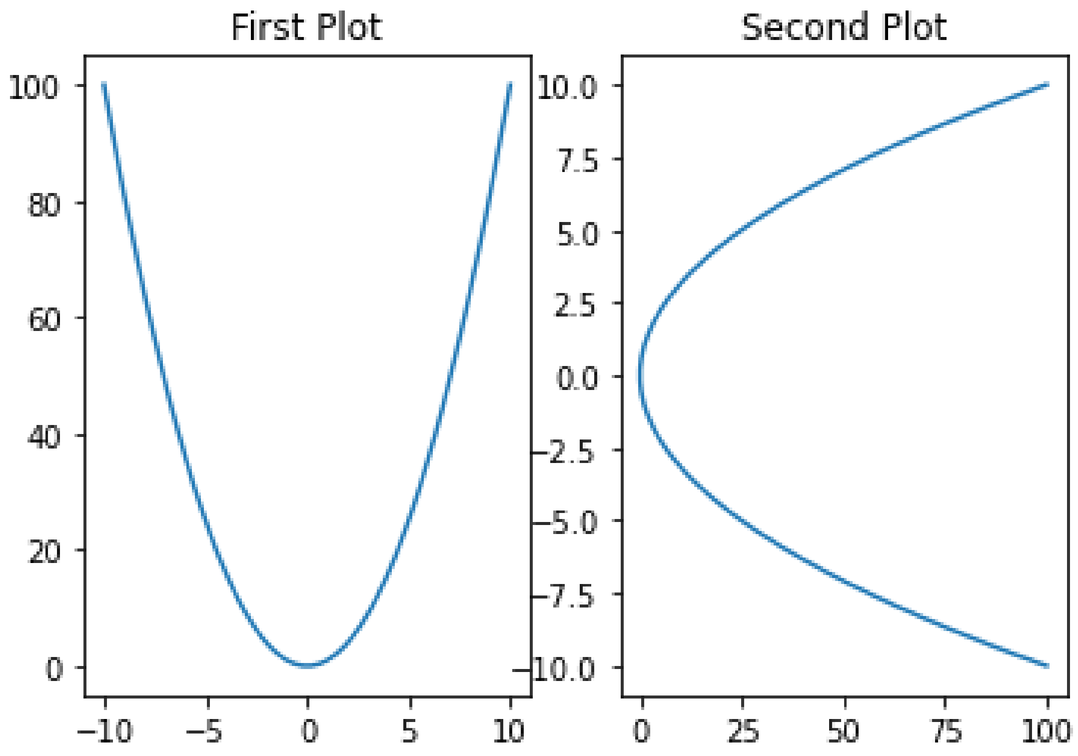
- **nrows**: the number of rows
- **ncols**: the number of columns
- **plot_number**: which refers to a specific plot in the Figure



Matplotlib

Multi-plots:

```
# Import module
from matplotlib import pyplot as plt
import numpy as np

x = np.linspace(-10, 10, 50)
y = x**2

plt.subplot(1, 2, 1)
plt.plot(x, y)
plt.title('First Plot')

plt.subplot(1, 2, 2)
plt.plot(y, x)
plt.title('Second Plot')
plt.show()
```



Matplotlib

Object oriented interface

- Best way to create plots
- Create Figure object and call methods of it
- Using the **.figure()** method to create

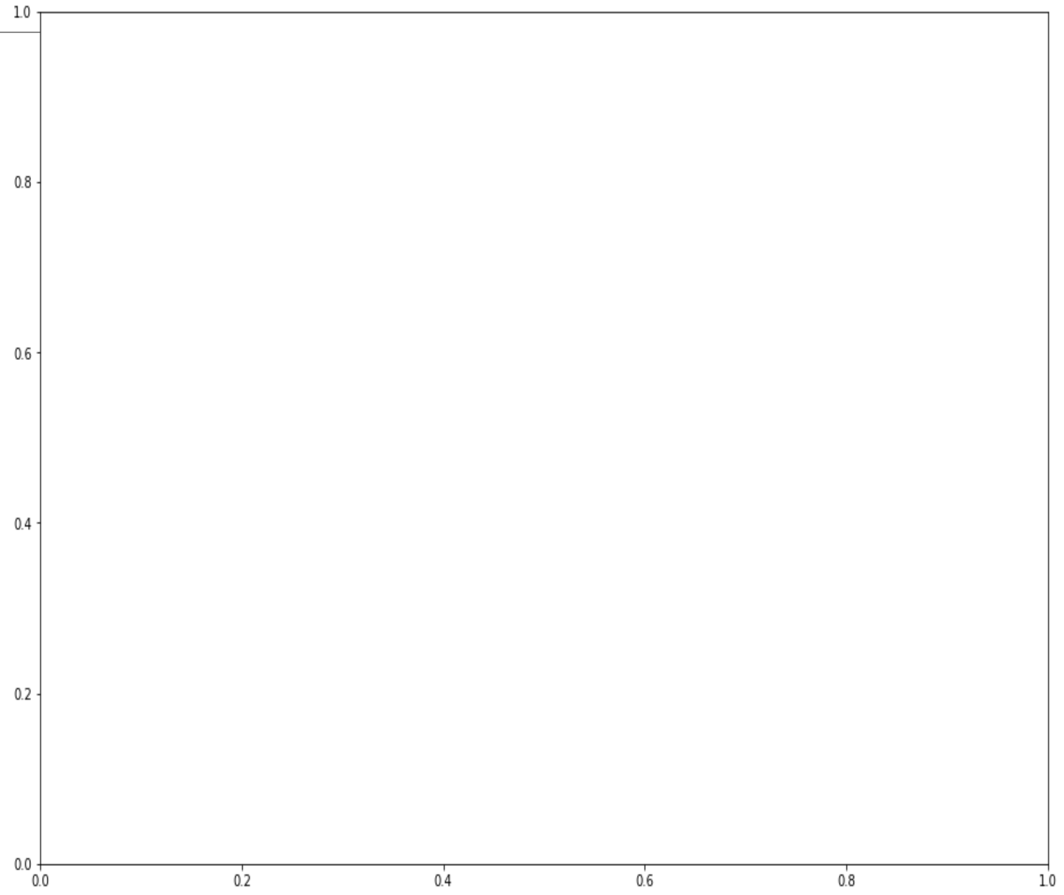
```python
# Import module
from matplotlib import pyplot as plt

fig = plt.figure()
```

Matplotlib

Object oriented interface

- Using the .**add_axes**([left, bottom, width, height]) method to create

```
# Import module
from matplotlib import pyplot as plt

fig = plt.figure()
ax = fig.add_axes([1, 2, 2, 2])
```

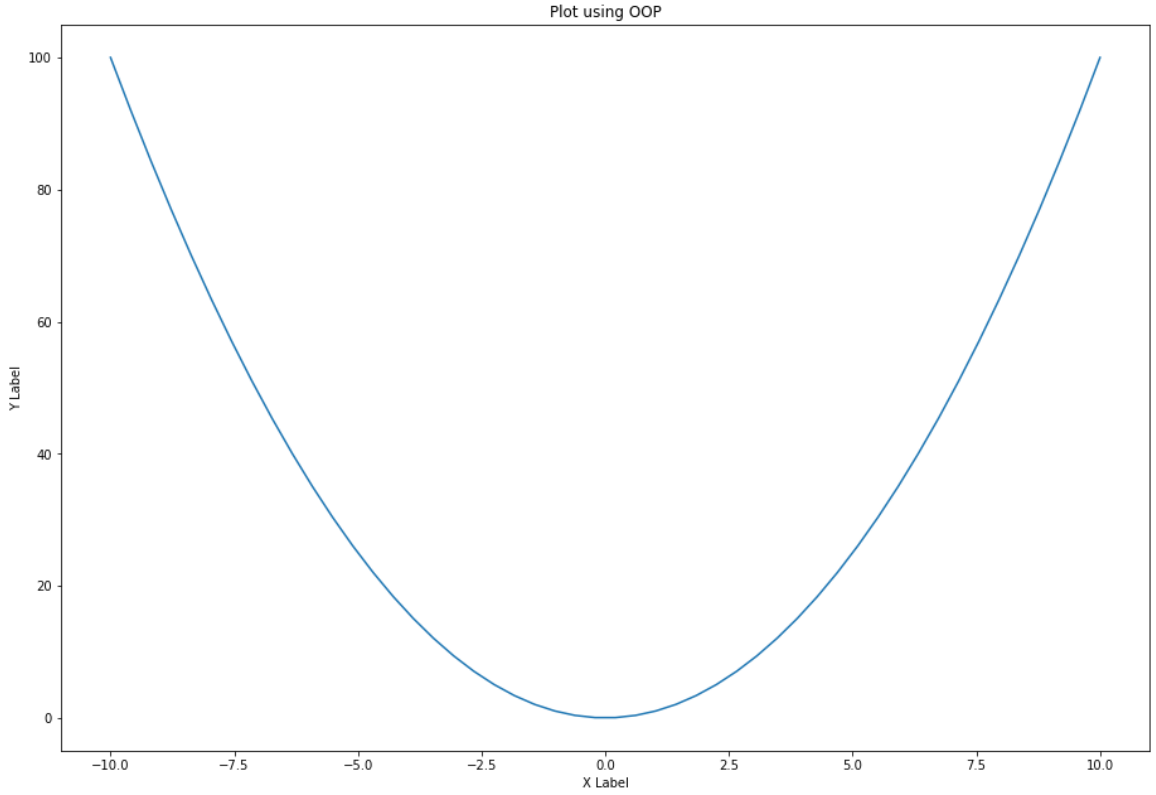


Matplotlib

Object oriented interface

- Plot values to axes

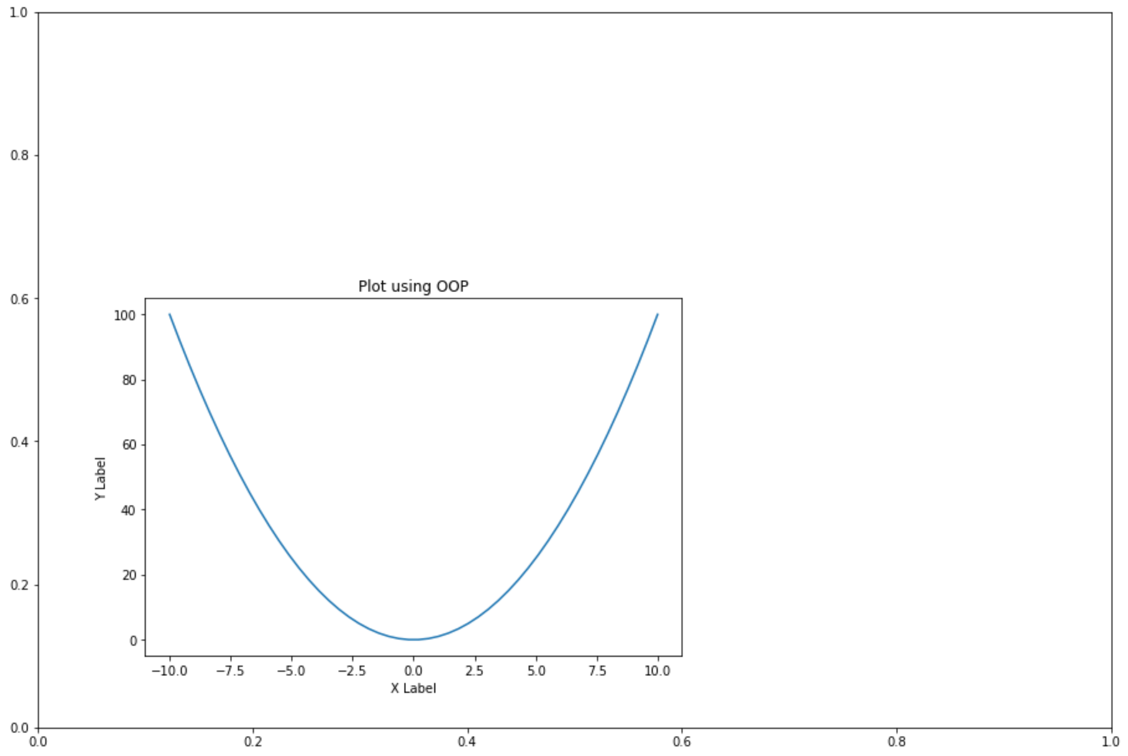
```
# Import module
from matplotlib import pyplot as plt
import numpy as np

x = np.linspace(-10, 10, 50)
y = x**2

fig = plt.figure()
ax = fig.add_axes([1, 2, 2, 2])
ax.plot(x, y)
ax.set_xlabel('X Label')
ax.set_ylabel('Y Label')
ax.set_title('Plot using OOP')
plt.show()
```



Matplotlib

Figure in Figure

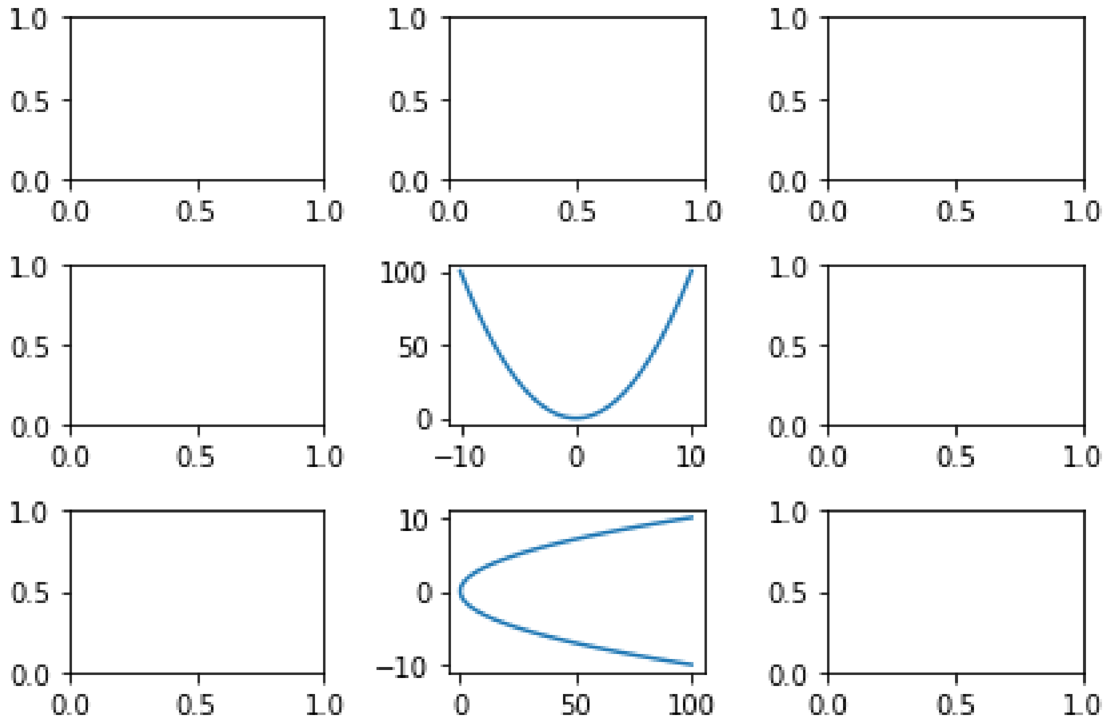
```
# Import module
from matplotlib import pyplot as plt
import numpy as np

x = np.linspace(-10, 10, 50)
y = x**2

fig = plt.figure()
axes1 = fig.add_axes([1, 2, 2, 2])
axes2 = fig.add_axes([1.2, 2.2, 1, 1])
axes2.plot(x, y)
axes2.set_xlabel('X Label')
axes2.set_ylabel('Y Label')
axes2.set_title('Plot using OOP')
plt.show()
```



Matplotlib

Matrix of subplot

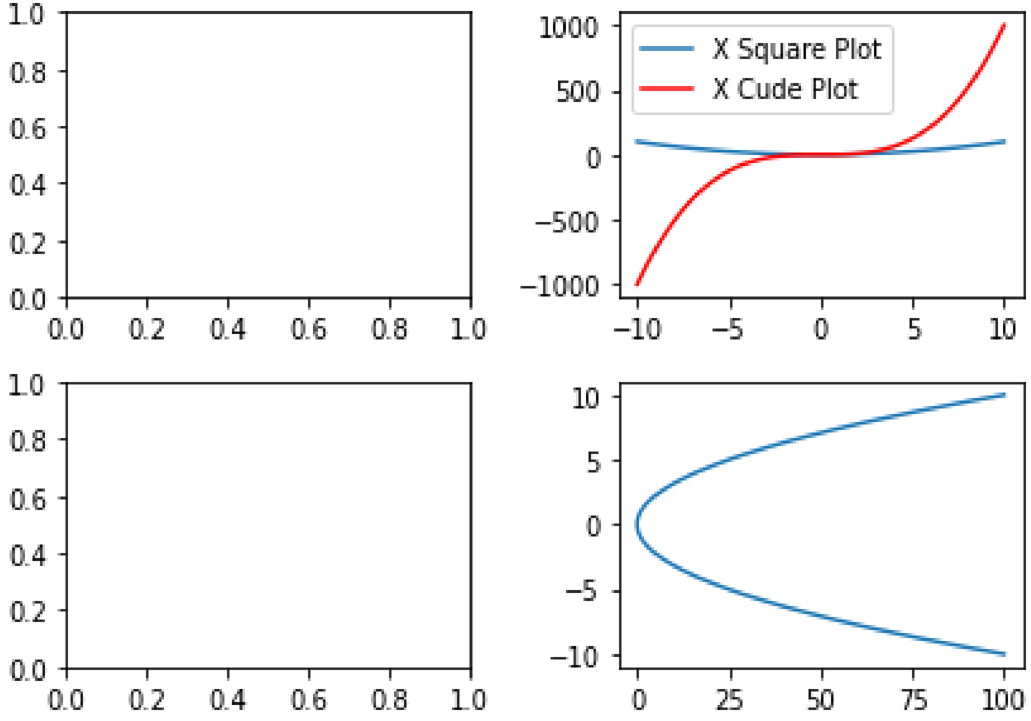
```
# Import module
from matplotlib import pyplot as plt
import numpy as np

x = np.linspace(-10, 10, 50)
y = x**2

fig, axes = plt.subplots(nrows=3, ncols=3)
axes[1, 1].plot(x, y)
axes[2, 1].plot(y, x)

# Automatically adjust subplot parameters to give specif
ied padding.
plt.tight_layout()
plt.show()
```



Matplotlib

Legend

```
# Import module
from matplotlib import pyplot as plt
import numpy as np
x = np.linspace(-10, 10, 50)
y = x**2
fig, axes = plt.subplots(nrows=2, ncols=2)
axes[0, 1].plot(x, y, label = 'X Square Plot')
axes[0, 1].plot(x, x**3, 'red' , label = 'X Cude Plot')
axes[0, 1].legend()
axes[1, 1].plot(y, x)
axes[0, 1].set_title('Legend')
# Automatically adjust subplot parameters to give
specified padding.
plt.tight_layout()
plt.show()
```



Scikit-Learn

- Machine learning library written in Python
- Simple and efficient
- Classical, **well-established machine learning algorithms**

Scikit-Learn - Algorithms

Supervised learning:

- Linear models (Ridge, Lasso, Elastic Net, ...)
- Support Vector Machines
- Tree-based methods (Classification/Regression Trees, Random Forests,...)
- Nearest neighbors
- Neural networks
- Gaussian Processes
- Feature selection

Scikit-Learn - Algorithms

Unsupervised learning:

- Clustering (KMeans, ...)
- Matric Decomposition (PCA, ...)
- Manifold Learning (Embeddings)
- Density estimation
- Outlier detection

Scikit-Learn

Model selection and evaluation:

- Cross-validation
- Grid-search
- Lots of metrics

THANK YOU !